

Percolation Weight and Thermodynamic Stability Across a 186-Compound Dataset

Campaign 2: Dataset Construction, VASP+LOBSTER Computation, and Statistical Analysis

Gilles Frapper
Professor, Theoretical Chemistry
IC2MP UMR 7285 CNRS, Université de Poitiers
gilles.frapper@univ-poitiers.fr

Report generated as part of the `viability` project

August 14, 2026

Contents

1	Objective and Problem Statement	2
2	Methodology	2
2.1	Dataset Selection	2
2.2	Bonding-Type Classification	2
2.3	VASP + LOBSTER Computation	2
2.4	Statistics	3
3	Workflow Diagram	3
4	Environment	3
5	Dataset	3
6	Results	4
6.1	Correlations (Spearman)	4
6.2	Reference Logistic Regression	4
6.3	Figures	5
7	Failures Found and Fixed	5
8	Compute Cost	6
9	Limits and Next Steps	6
10	Conclusion	6
A	Compound List	7

1 Objective and Problem Statement

The first report (6 toy compounds) established that the percolation descriptor computed by `percolation_path.py` diverges sharply from classic ICOHP aggregates (sum, mean, min, max), but without a sample large enough to statistically test whether that difference is *useful*: does the percolation descriptor carry predictive information about thermodynamic stability (`energy_above_hull`) that the classic aggregates don't already carry?

This second campaign builds a 186-compound dataset (180 new + the 6 pilot compounds) split into three families:

- **exp_stable**: experimental (ICSD-referenced) compounds on the Materials Project thermodynamic convex hull;
- **exp_metastable**: experimental metastable compounds ($0 < E_{\text{hull}} \leq 100$ meV/atom);
- **theo_metastable**: theoretical (never experimentally observed) metastable compounds ($0 < \bar{E}_{\text{hull}} \leq 200$ meV/atom).

The `exp_metastable` / `theo_metastable` split serves as a proxy for kinetic persistence: an experimentally observed metastable compound has, by definition, survived long enough to be synthesized and characterized, unlike a theoretical compound of comparable thermodynamic metastability.

2 Methodology

2.1 Dataset Selection

Materials Project query (`mp_dataset/select_campaign.py`): unary and binary chemical systems, non-magnetic ($|\text{total magnetization}| \leq 0.01 \mu_B/\text{formula unit}$ — MP's categorical `magnetic_ordering` field is "Unknown" for ~98% of entries and would have excluded nearly every genuinely non-magnetic compound), no f-block elements, ≤ 20 atoms/cell, 60 compounds per family with a 2-entries-per-chemical-system cap to ensure elemental diversity rather than many polymorphs of the same system.

2.2 Bonding-Type Classification

Bond type (ionic / covalent / metallic) is computed by `fetch_candidates.py`'s `classify()` function (**reused unchanged**, per the campaign's instructions), from `is_metal` and elemental composition. This is a secondary analysis axis here, not the primary sampling axis (unlike the campaign's originally proposed plan, replaced at the user's request with the stable/metastable scheme above).

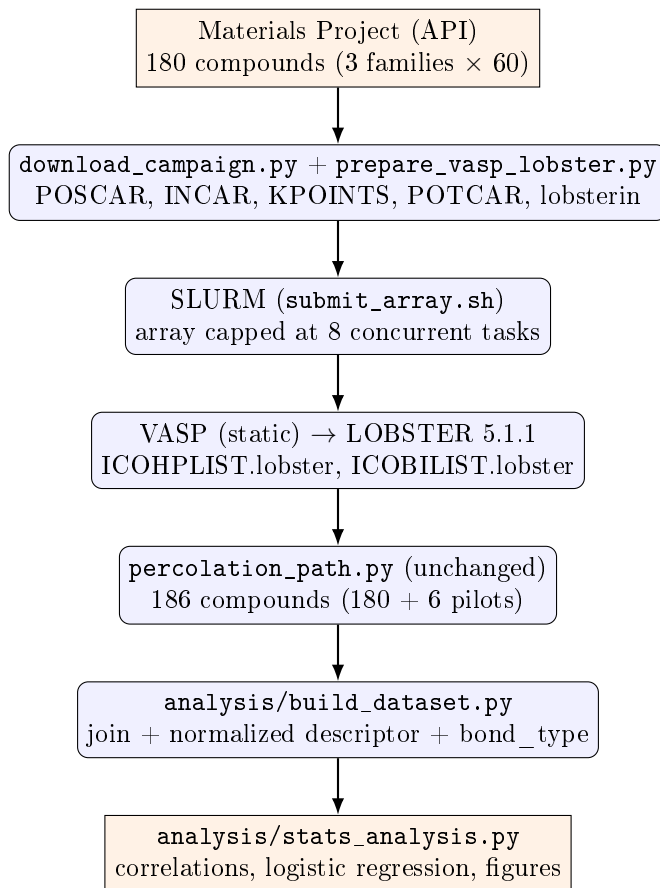
2.3 VASP + LOBSTER Computation

Same template as the pilot campaign (static run, `ISYM=-1`, `LASPH=.TRUE.`, PAW-PBE potentials recommended by `pymatgen.io.vasp.sets.MPRelaxSet`), with two improvements: `NBANDS` is no longer a fixed value but derived per compound from the number of local orbitals in the LOBSTER basis (`Lobsterin.write_INCAR`), and basis-function determination goes through `Lobsterin.get_basis(potcar_symbols=...)` rather than direct POTCAR-file parsing (whose internal pymatgen validity check fails on several POTCARs unrecognized by hash in the local mirror).

2.4 Statistics

Spearman correlation (no reason to expect a linear relationship) between percolation weight (raw and normalized by $|\text{ICOHP}_{\min}|$) and `energy_above_hull`, overall and per bond type; same for the classic ICOHP aggregates, for direct comparison; a reference logistic regression (stable vs. metastable, 2–3 features, 5-fold cross-validation) for a reference AUC. **No SISSO, no symbolic regression**: premature with a single candidate descriptor at this sample size (see §9).

3 Workflow Diagram



4 Environment

Same as the pilot campaign (Yargla, VASP 6.5.0, LOBSTER 5.1.1, Python 3.11 `.venv`), extended with: `mp-api`, `pandas`, `scipy`, `scikit-learn`, `matplotlib` (see `requirements.txt`).

5 Dataset

Family	n	E_{hull} range (eV/atom)
exp_stable	63	0.0000 (by construction)
exp_metastable	63	0.0005 – 0.0907
theo_metastable	60	0.0013 – 0.1991

Bond type (<code>classify()</code>)	<i>n</i>
metallic	88
covalent	23
ionic	9
unclassified	66

35% of compounds (66/186) fall outside `classify()`'s three categories — mostly transition-metal pnictides/chalcogenides and hydrogen-containing compounds, a real limitation of the reused heuristic, documented but not fixed here.

6 Results

6.1 Correlations (Spearman)

Group	Metric	<i>n</i>	ρ	<i>p</i>
all	percolation weight (raw)	186	0.064	0.383
all	percolation weight (normalized)	186	0.071	0.339
metallic	percolation weight (raw)	88	0.191	0.075
metallic	percolation weight (normalized)	88	0.222	0.038
ionic*	percolation weight (raw)	9	−0.402	0.284
covalent	percolation weight (raw)	23	0.165	0.453
all	ICOHP sum	186	0.032	0.670
all	ICOHP min	186	0.093	0.209
metallic	ICOHP sum	88	0.223	0.037
metallic	ICOHP min	88	0.198	0.065

* *n* < 15: do not over-interpret.

In the one subgroup where any result reaches nominal significance (metallic, *n* = 88), the normalized percolation weight (ρ = 0.222, *p* = 0.038) and the ICOHP sum (ρ = 0.223, *p* = 0.037) are statistically indistinguishable in strength. **The percolation descriptor does not demonstrate additional predictive power over the classic aggregates on this dataset** — a more precise answer than the previous (6-compound) report's "qualitatively different" framing, and a more directly actionable one.

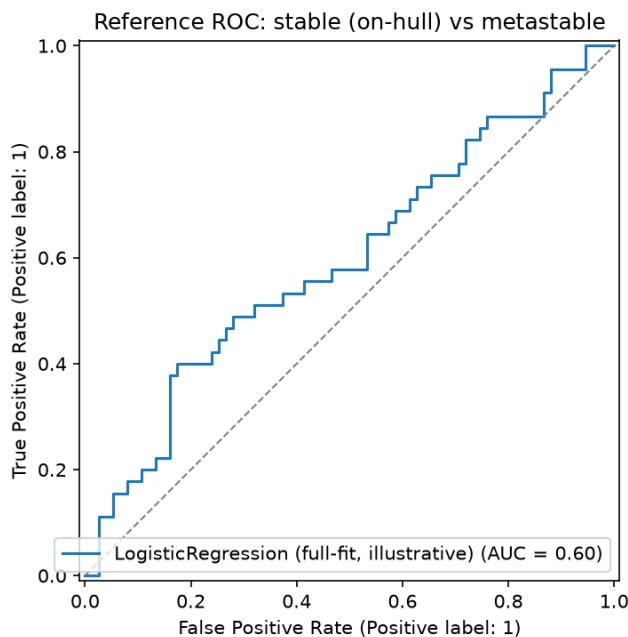
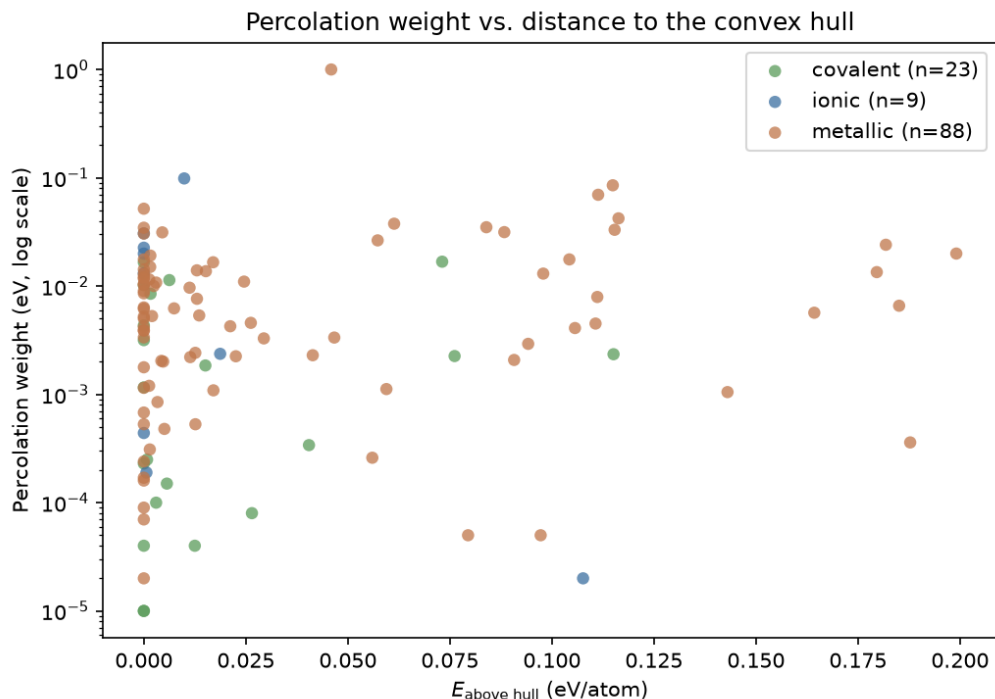
6.2 Reference Logistic Regression

Stable (on-hull) vs. metastable, features: normalized percolation weight, mean ICOHP, bond-type dummies (*n* = 120 after dropping unclassified compounds).

AUC (5-fold cross-validation)	0.496 ± 0.141
per-fold AUC	0.556, 0.430, 0.252, 0.637, 0.607

AUC $\approx$ 0.5: chance level. The wide spread across folds (0.25–0.64) is itself informative: it indicates the absence of a stable, learnable signal at this sample size, not merely a "weak but real" one.

6.3 Figures



7 Failures Found and Fixed

Per the instruction ("every job failure must be logged with its probable cause"), 5 of the 178 SLURM-array-submitted jobs failed on the first pass, all diagnosed and fixed at the source:

- **Al12W, BW, WO2, TeW, TiW** (all tungsten-containing): the local mirror's `W_sv` POTCAR is missing its `LEXCH` field entirely (confirmed by direct inspection — present in every other variant checked, including plain `W`), which VASP reads as an incompatible exchange-correlation functional and refuses to run. Fixed by replacing the `W_sv` override with plain `W` (valid PBE, verified).

- **A112W** failed again after that fix, this time with a VASP SIGSEGV — exactly the known issue flagged in the mission brief (`ulimit -s unlimited` before `vasp_std`), present in `submit_array.sh` but forgotten in `prepare_vasp_lobster.py`'s per-compound SLURM template. Fixed; recovered on retry.

Final success rate: **186/186**.

8 Compute Cost

	VASP	LOBSTER
Mean (s)	124	397
Median (s)	72	145
Max (s)	1440	6675
Sequential total	6.4 h	19.1 h

Actual elapsed wall-clock time (8-way concurrency): ~ 3.5 h, within the 6 h overnight budget.

9 Limits and Next Steps

- **Sample size:** 186 overall, but 88/23/9 once split by bond type — the "ionic" correlations ($n = 9$) are not interpretable.
- **Primitive vs. conventional cell** (carried over from the pilot report): not addressed here. Most likely explanation for the weak observed correlations — the strongest bond can sit entirely inside the primitive cell and contribute to no non-contractile cycle at all, biasing the measured "weakest link."
- **No true dynamical "unstable" class:** `theo_metastable` is a proxy (never synthesized), not a phonon-calculation result.
- **classify() coverage:** 35% of compounds unclassified.
- **Why not SISSO yet:** a single candidate descriptor (the percolation weight, in two algebraically related variants), compared against aggregates that are themselves functions of the same underlying ICOHP values — no meaningful combinatorial search space yet. Justifying SISSO would require: (a) several conceptually novel candidate descriptors, (b) per-stratum $n \geq 50$ –100, (c) a clearer signal than $AUC \approx 0.5$.

10 Conclusion

The 186-compound dataset was built, computed (VASP+LOBSTER), and analyzed end to end, with two compute failures diagnosed and fixed at the source. The headline result is a *negative* but quantified and honest one: at this sample size, the percolation descriptor shows no significant correlation with thermodynamic stability, and demonstrates no advantage over classic ICOHP aggregates. The priority actions (§9) — especially conventional-cell reprocessing and a targeted extension of the metallic subsample — are likely to move this result before a SISSO-class tool is warranted.

Source code: <https://github.com/John-Akapulco/viability>

Data: `analysis/percolation_vs_hull.csv`

Detailed report: `analysis/REPORT.md`

A Compound List

The 186 compounds in the dataset, grouped by category (family), with their Materials Project identifier, PBE thermodynamic hull distance (`energy_above_hull`), and the minimum ICOHP percolation weight (`icohp_percolation_weight_min`, this work’s main metric). Auto-generated from `analysis/percolation_vs_hull.csv` by `report/gen_appendix.py`.

Table 1: Complete list of the 186 compounds, grouped by category, with their Materials Project identifier, PBE hull distance, and minimum ICOHP percolation weight.

Formula	mp-id	Bonding	E_{hull} (eV/at)	Percolation weight (eV)
Experimental, stable (hull) ($n = 63$)				
As2S3	mp-1189572	covalent	0.0000	1.00e-05
AsF3	mp-28027	covalent	0.0000	0.00e+00
BI3	mp-23189	covalent	0.0000	1.00e-05
BW	mp-7832	—	0.0000	1.90e-04
BaAu2	mp-30363	metallic	0.0000	5.19e-02
BaCl2	mp-568662	ionic	0.0000	1.31e-02
BaH2	mp-23715	—	0.0000	2.40e-04
Be2Cu	mp-2031	metallic	0.0000	1.15e-03
Be2Nb3	mp-11272	metallic	0.0000	6.36e-03
Be2V	mp-11281	metallic	0.0000	6.80e-04
CaAl4	mp-1749	metallic	0.0000	1.03e-02
CaSn	mp-2450	metallic	0.0000	5.23e-03
CdPd	mp-1696	metallic	0.0000	9.00e-05
CoB	mp-20857	—	0.0000	6.00e-05
Cs2As3	mp-15557	—	0.0000	2.02e-02
FeS	mp-505531	—	0.0000	3.05e-03
Ge2Os	mp-10032	metallic	0.0000	1.21e-02
HfTc2	mp-1095669	metallic	0.0000	1.77e-02
Hg4Pt	mp-936	metallic	0.0000	3.36e-03
HgF2	mp-8177	—	0.0000	8.95e-03
InBr	mp-22870	covalent	0.0000	1.02e-02
KI	mp-22898	ionic	0.0000	2.26e-02
KN3	mp-827	—	0.0000	2.80e-04
Li2Se	mp-2286	—	0.0000	1.16e-02
LiB	mp-1001835	—	0.0000	2.91e-03
Mg2Ni	mp-2137	metallic	0.0000	1.60e-04
MgP4	mp-384	—	0.0000	0.00e+00
MnAl12	mp-1104165	metallic	0.0000	1.42e-02
MoSe2	mp-1634	covalent	0.0000	1.16e-03
NaS	mp-2400	—	0.0000	0.00e+00
Na	mp-10172	—	0.0000	1.97e-03
Nb3Ir	mp-1458	metallic	0.0000	9.02e-03
NbGa3	mp-1973	metallic	0.0000	8.58e-03
PI2	mp-29443	covalent	0.0000	2.30e-04
PbSe	mp-2201	covalent	0.0000	1.66e-02
RbI	mp-22903	ionic	0.0000	2.00e-02
Sc5Ge3	mp-17190	metallic	0.0000	5.30e-04
ScAg2	mp-1018128	metallic	0.0000	1.70e-04
ScTe	mp-10026	—	0.0000	7.12e-03
SnPt	mp-19856	metallic	0.0000	1.29e-02
SrO2	mp-2697	ionic	0.0000	4.40e-04
SrSi	mp-2661	metallic	0.0000	7.00e-05
Ta5Ge3	mp-17593	metallic	0.0000	3.46e-02

Table 1 (continued)

Formula	mp-id	Bonding	E_{hull} (eV/at)	Percolation weight (eV)
TaP	mp-1067587	—	0.0000	3.14e-02
TaSb2	mp-11697	metallic	0.0000	1.17e-02
Te2Pd	mp-782	—	0.0000	1.44e-02
Te2Ru	mp-267	covalent	0.0000	4.34e-03
TeO2	mp-8377	covalent	0.0000	4.00e-05
Ti3In	mp-866046	metallic	0.0000	5.01e-03
Ti3Pt	mp-2339	metallic	0.0000	1.78e-03
TiBe12	mp-1104067	metallic	0.0000	2.40e-04
TiNi	mp-1048	metallic	0.0000	3.91e-03
TiO	mp-1071163	—	0.0000	6.40e-04
TiSi2	mp-1077503	metallic	0.0000	6.18e-03
TiCl	mp-571079	—	0.0000	7.17e-03
VNi2	mp-11531	metallic	0.0000	3.88e-03
WO2	mp-1524394	—	0.0000	2.20e-04
Zr2Au	mp-2348	metallic	0.0000	3.06e-02
Zr5Pb3	mp-681992	metallic	0.0000	2.00e-05
ZrMo2	mp-2049	metallic	0.0000	1.05e-02
Si	mp-149	covalent	0.0000	3.17e-03
NaCl	mp-22862	ionic	0.0000	3.07e-02
AlNi	mp-1487	metallic	0.0000	4.19e-03
Experimental, metastable ($n = 63$)				
CdI2	mp-567259	—	0.0005	2.38e-02
MgCl2	mp-570259	ionic	0.0006	1.90e-04
C	mp-169	covalent	0.0008	2.50e-04
ZrBr	mp-1065846	—	0.0008	7.88e-03
Li3Hg	mp-1646	metallic	0.0013	1.20e-03
Ta7Co6	mp-1104548	metallic	0.0015	3.10e-04
CaGa4	mp-2461	metallic	0.0016	1.91e-02
InCl	mp-571636	covalent	0.0016	8.50e-03
NiB	mp-14019	—	0.0018	4.10e-04
MgH2	mp-23711	—	0.0019	1.00e-05
Sn7Ir5	mp-20466	metallic	0.0024	1.00e-02
NaN3	mp-1066400	—	0.0029	7.00e-04
ClF3	mp-556767	—	0.0030	0.00e+00
SiS2	mp-1095294	covalent	0.0030	1.00e-04
Sr5Sn3	mp-17720	metallic	0.0030	1.08e-02
Te2Ir	mp-1551	—	0.0030	2.80e-03
Li13Sn5	mp-30769	metallic	0.0033	8.50e-04
H3N	mp-643432	covalent	0.0038	0.00e+00
Cs2Se	mp-1011696	—	0.0042	2.62e-02
In3Ir	mp-630976	metallic	0.0043	2.04e-03
HI	mp-697025	—	0.0044	0.00e+00
Rb2Hg7	mp-31474	metallic	0.0045	3.14e-02
PbAu2	mp-19871	metallic	0.0047	2.01e-03
BeCu	mp-2323	metallic	0.0050	4.80e-04
SiO2	mp-546794	covalent	0.0056	1.50e-04
IF7	mp-27988	—	0.0063	0.00e+00
K	mp-10157	—	0.0069	8.07e-02
Al3Os2	mp-16521	metallic	0.0074	6.23e-03
Nb2O5	mp-1595	—	0.0075	1.00e-04
LiBr	mp-23259	ionic	0.0099	9.88e-02
ZnAg	mp-1912	metallic	0.0112	9.67e-03
MgPd3	mp-7753	metallic	0.0114	2.21e-03
AsPd2	mp-1080831	—	0.0118	3.50e-04

Table 1 (continued)

Formula	mp-id	Bonding	E_{hull} (eV/at)	Percolation weight (eV)
GeSe2	mp-10074	covalent	0.0125	4.00e-05
Re3Ge7	mp-29141	metallic	0.0126	2.42e-03
Cu3Ge	mp-19724	metallic	0.0126	5.30e-04
Ti3Ge	mp-1189044	metallic	0.0130	1.40e-02
ZrGa3	mp-30685	metallic	0.0130	7.64e-03
SrAl2	mp-1638	metallic	0.0136	5.36e-03
NiS	mp-1547	—	0.0142	3.00e-04
Al12W	mp-11227	metallic	0.0152	1.37e-02
CsH	mp-632319	—	0.0161	2.66e-02
CaGa2	mp-992	metallic	0.0170	1.66e-02
NiHg4	mp-570835	metallic	0.0171	1.09e-03
SiRh	mp-2287	metallic	0.0246	1.10e-02
BaSn5	mp-571353	metallic	0.0262	4.58e-03
PS	mp-612	covalent	0.0265	8.00e-05
AsRh	mp-22079	—	0.0269	1.93e-03
Te2Au	mp-567525	—	0.0274	1.62e-02
Be3N2	mp-6977	—	0.0315	2.00e-05
Y2O3	mp-558573	—	0.0342	5.00e-05
AgO	mp-1079720	—	0.0423	4.00e-05
FeTe2	mp-1102002	—	0.0449	5.20e-04
H2O	mp-2739191	—	0.0494	0.00e+00
CdO2	mp-2310	—	0.0562	9.00e-05
Mo3Se4	mp-21021	—	0.0609	1.84e-03
PbSe	mp-22009	covalent	0.0731	1.68e-02
Be12Pd	mp-12666	metallic	0.0795	5.00e-05
Sr2As	mp-15697	—	0.0800	2.06e-03
AsRh2	mp-1102364	—	0.0806	4.63e-03
ZrRh	mp-2808	metallic	0.0839	3.50e-02
K2S	mp-559278	—	0.0873	1.07e-02
Ge4Rh	mp-1104286	metallic	0.0907	2.08e-03
Theoretical, metastable ($n = 60$)				
Li3Ag	mp-976408	metallic	0.0013	1.15e-02
Y3Co	mp-1189329	metallic	0.0016	1.50e-02
TiNi	mp-1067248	metallic	0.0020	5.29e-03
TaP	mp-1187244	—	0.0033	3.59e-02
Sr3As4	mp-678007	—	0.0062	4.47e-03
Ga2Se3	mp-1224809	covalent	0.0063	1.14e-02
MnN	mp-1009130	covalent	0.0151	1.85e-03
BeCl2	mp-1190080	ionic	0.0187	2.37e-03
MgSn	mp-1094801	metallic	0.0212	4.26e-03
TiW	mp-1216621	metallic	0.0225	2.25e-03
SrTi3	mp-1206636	metallic	0.0294	3.29e-03
IrCl3	mp-729507	—	0.0301	2.72e-03
Ta4C3	mp-1218000	—	0.0335	1.44e-03
Sc2O3	mp-755313	—	0.0364	5.10e-04
SnBr2	mp-978985	covalent	0.0405	3.40e-04
Sn7Pd5	mp-1219006	metallic	0.0414	2.30e-03
Tl4Sn	mp-1216560	metallic	0.0459	1.00e+00
LiTi3	mp-1185253	metallic	0.0467	3.35e-03
PtCl2	mp-1219748	—	0.0490	5.04e-03
HCl	mp-684609	—	0.0536	2.50e-04
MgSn	mp-1094669	metallic	0.0560	2.60e-04
K5As4	mp-684799	—	0.0564	8.80e-03
KSr3	mp-1185151	metallic	0.0573	2.65e-02

Table 1 (continued)

Formula	mp-id	Bonding	E_{hull} (eV/at)	Percolation weight (eV)
NbS2	mp-774873	—	0.0576	1.20e-04
YAg3	mp-1187811	metallic	0.0594	1.12e-03
B3Ir2	mp-1228647	—	0.0609	2.80e-04
TiMo2	mp-1216675	metallic	0.0613	3.77e-02
Cr2C	mp-1226378	—	0.0625	2.08e-03
ZrO2	mp-775909	—	0.0627	1.20e-04
Zn	mp-2647117	—	0.0717	1.04e-02
IN2	mp-1097011	—	0.0730	4.70e-04
CoSe2	mp-1390196	—	0.0734	8.40e-04
GeS	mp-12910	covalent	0.0762	2.26e-03
ZrTi	mp-1215236	metallic	0.0883	3.15e-02
VCo3	mp-1095057	metallic	0.0942	2.93e-03
Tl3Ga	mp-1187539	metallic	0.0972	5.00e-05
CaGe2	mp-643542	metallic	0.0979	1.30e-02
NbRh2	mp-1220414	metallic	0.1043	1.76e-02
TcW	mp-1217337	metallic	0.1056	4.10e-03
MgF2	mp-560236	ionic	0.1076	2.00e-05
VN	mp-1017532	—	0.1080	4.00e-05
Na2Cl	mp-990084	—	0.1105	3.76e-02
ZnPb3	mp-1187949	metallic	0.1107	4.51e-03
Sr3Sn	mp-1187136	metallic	0.1111	7.94e-03
TaRe	mp-1217894	metallic	0.1113	6.98e-02
Al4Si	mp-1228120	metallic	0.1149	8.54e-02
B6Pb	mp-1096825	covalent	0.1151	2.35e-03
ScTc3	mp-867262	metallic	0.1154	3.31e-02
CrMo	mp-1226200	metallic	0.1163	4.23e-02
BPd5	mp-1228665	—	0.1325	2.42e-02
Zr3Be	mp-1188016	metallic	0.1430	1.05e-03
Na3Cl	mp-1064484	—	0.1456	2.55e-03
SiAu3	mp-1186998	metallic	0.1643	5.68e-03
InSe2	mp-1232036	—	0.1733	1.64e-03
TiSi2	mp-570991	metallic	0.1796	1.35e-02
BeAu3	mp-983539	metallic	0.1818	2.41e-02
Zr3Ge	mp-1188039	metallic	0.1851	6.59e-03
AlSb	mp-1023918	metallic	0.1878	3.60e-04
Mg15B	mp-1023515	—	0.1943	2.80e-04
HfMo	mp-1224314	metallic	0.1991	2.00e-02